



Interim Progress Report
*A Survey of Deep Learning-Based Reconstruction Methods
for Quantum Imaging Systems*
PARIMANA Summer Internship Program 2026 — Qmet Tech
Foundation



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1. Introduction

This report summarises work completed during the first three weeks of the PARIMANA Summer Internship at Qmet Tech Foundation. The project aims to produce a survey of deep learning methods applied to image reconstruction in quantum optical systems, covering quantum ghost imaging, compressed sensing with entangled photons, sub-shot-noise imaging, and hybrid classical-quantum frameworks.

2. Work Completed

Literature Review. A systematic search across IEEE Xplore, arXiv, Google Scholar, and Optica Publishing yielded approximately 45 candidate papers (1995–2025). A three-axis taxonomic framework was developed to classify papers by imaging modality, deep learning architecture, and reconstruction task. The ghost imaging domain has been reviewed in depth, tracing the field from Pittman et al. (1995) through the first deep learning application by Lyu et al. (2017), the CNN approach of He et al. (2018), and the end-to-end framework of Wang et al. (2019). Compressed sensing methods based on Katz et al. (2009) and sub-shot-noise imaging following Brida et al. (2010) have been partially reviewed.

Software Development. A Python/PyTorch framework has been developed for empirical validation. It includes: (i) a computational ghost imaging simulator supporting configurable illumination patterns, noise, and correlation-based reconstruction; (ii) three deep learning reconstruction models (fully connected network, CNN, and U-Net) corresponding to the principal architectures identified in the survey; and (iii) a training and evaluation pipeline with PSNR/SSIM metrics. Preliminary experiments with the fully connected network at 12.5% sampling ratio achieved a validation PSNR of approximately 22 dB, consistent with published results.

3. Survey Manuscript Status

The survey manuscript currently stands at approximately 60–70% completion. The introduction, background, and ghost imaging sections are substantially complete (approximately seven two-column pages). The compressed sensing, sub-shot-noise, hybrid architectures, discussion, and conclusion sections are in draft form. The bibliography contains 22 verified references.

4. Summary

The project is progressing on schedule. The literature review covers ghost imaging and compressed sensing in depth, with remaining domains under active review. A working software framework has been developed and validated. The remaining period will focus on completing the outstanding review sections, running comprehensive experiments across all architectures, and finalising the manuscript.

References: T.B. Pittman et al., *Phys. Rev. A* **52**, R3429 (1995); M. Lyu et al., *Sci. Rep.* **7**, 17865 (2017); Y. He et al., *Sci. Rep.* **8**, 6469 (2018); F. Wang et al., *Opt. Express* **27**, 25560 (2019); O. Katz et al., *Appl. Phys. Lett.* **95**, 131110 (2009); G. Brida et al., *Nat. Photon.* **4**, 227 (2010); B.I. Erkmen & J.H. Shapiro, *Adv. Opt. Photon.* **2**, 405 (2010); G. Barbastathis et al., *Optica* **6**, 921 (2019).